

dat

1.Explaining Tourist Revisit Intention Using Natural Language

Datasets:Primary dataset: 75,000 TripAdvisor hotel reviews from Cyprus (2010–2019).
Secondary dataset: 515,000 Booking.com hotel reviews for training the classifier.

Method:

here researchers used Natural Language Processing (NLP), Word2Vec, negation detection, Structural Topic Modeling (STM), XGBoost classifier, and SHAP explainable AI.

2.An Unsupervised Aspect-Sentiment Model for Online Reviews

Dataset: Reviews: More than 50,000 Citysearch New York restaurant reviews. Evaluation dataset:

3,400 manually annotated sentences. Netbook Reviews: 1,086 Amazon netbook reviews to test domain independence.

Method:

The authors proposed a fully unsupervised aspect-sentiment model. They used Local LDA (Latent Dirichlet Allocation) for aspect extraction, automatic seed word generation using negation, graph-based label propagation, and aspect-specific sentiment analysis.

3.Opinion Digger: An Unsupervised Opinion Miner from Unstructured Product Reviews

datasets:2,483 (about 2.5K) Epinions reviews from 5 product categories: Camcorder, Cellular Phone, Digital Camera, DVD Player, and MP3 Player.

Method:

The authors proposed Opinion Digger, an unsupervised method. It uses POS tagging, Apriori algorithm, GSP pattern mining, WordNet, and KNN (2-NN) for aspect extraction and aspect rating.

4.A Survey on Aspect-Based Sentiment Analysis: Tasks, Methods, and Challenges

Dataset:The paper reviews many benchmark datasets, including SemEval-2014, SemEval-2015, SemEval-2016, Twitter, Restaurant, Laptop, and Movie review datasets.

methods: The authors reviewed and compared machine learning, deep learning, transformer-based models (BERT), lexicon-based methods, and hybrid approaches for ABSA.

5.Aspect-Based Sentiment Analysis Using BERT

Dataset:

SemEval-2014 Restaurant, SemEval-2014 Laptop, SemEval-2015 Restaurant, and SemEval-2016 Restaurant datasets.

Method:

The authors used BERT with two methods: sentence-pair classification and single-sentence classification. They fine-tuned the pre-trained BERT model for aspect sentiment classification.

6.Implicit Polarity and Implicit Aspect Recognition in Opinion Mining

Dataset:Chinese hotel reviews from Booking.com containing 144,158 positive and 113,844 negative reviews from 20,973 hotels in 49 international cities.

Method:The authors collected Chinese hotel reviews, built an implicit opinion corpus, and used Bag of Words (BOW), Word2Vec, SVM (Linear & RBF), and CNN for aspect and polarity classification.

7 Feature Extraction from Customer Reviews Using Enhanced Rules

Dataset:

The study uses 5 electronic product review datasets from Hu and Liu (2004): 2 digital cameras, 1 cell phone, 1 MP3 player, and 1 DVD player. It uses 3,945 review sentences, including 1,726 labelled sentences.

Method:

This paper used 41 sequential pattern-based rules to extract explicit product features. It includes 25 old rules and 16 new rules. The method also uses text preprocessing, opinion lexicons, and POS tagging with NLTK and Flair.

8

Dataset:

Source: Chinese online auto forum (Autohome). • Product: Buick Envision passenger car. • Dataset size: 2,986 online reviews from 2016–2021.

Method:

online customer reviews. Used NLP to find product features and customer sentiments. Created a customer network based on similar opinions. Used Graph Autoencoder (GAE) to reduce complex data. Used K-means clustering to group customers.

9.A Classification-based Approach for Implicit Feature Identification

Dataset:

Source: Amazon.cn online reviews. Domains: Cell phones and digital cameras.

Cell phones: 2,694 reviews and 8,305 sentences. Digital cameras: 1,389 reviews and 4,455 sentences. • Total: 4,083 reviews and 12,760 sentences, including 6,050 explicit features and 2,247 implicit features.

Method

Extracted explicit feature-opinion pairs using Chinese dependency grammar rules. Grouped feature-opinion pairs that shared the same opinion word. Built training documents from explicit sentences. Changed implicit feature identification into a text classification problem. Used a Topic-Feature-Centroid Classifier with TF-IDF-like weighting and cosine similarity to identify implicit features.

10.Detection of Multiple Implicit Features per Sentence in Consumer Review Data.

Method:

Extended an earlier supervised method to identify multiple implicit features in one sentence. • Used logistic regression with four variables: number of nouns, adjectives, commas, and the word "and". • Used a trained threshold to decide whether a sentence may contain multiple implicit features. • Used another threshold based on feature co-occurrence scores to select the features. • Applied POS filtering using nouns and adjectives (NN + JJ) to remove irrelevant words.

Dataset:

Source: Restaurant review dataset by Ganu et al. (2009). • Size: 3,044 sentences. • Main feature categories: food, service, ambience, and price. • 52.6% of sentences had 1 implicit feature, 12.4% had 2, 2.3% had 3, 0.1% had 4, and 32.7% had no implicit feature.

11.Clustering for Simultaneous Extraction of Aspects and Features from Reviews

Method: Proposed CAFE (Clustering for Aspect and Feature Extraction), an unsupervised method that extracts features and groups them into aspects at the same time. • Used POS tagging to identify nouns/noun phrases as explicit features and adjectives/verbs as possible implicit features. • Used frequent candidates to create initial clusters and reduce noise. • Combined general word meaning, word co-occurrence, and statistical similarity to measure the relationship between features. • Used clustering rules to merge similar features into aspect groups.

Dataset:

Cell Phone Reviews: 500 reviews, 46 aspects, 419 explicit features, and 339 implicit features. • GPS Reviews: 500 reviews, 37 aspects, 637 explicit features, and 492 implicit features. • TV Reviews: 500 reviews, 34 aspects, 485 explicit features, and 277 implicit features.

12.Unsupervised Extraction of Popular Product Attributes from E-Commerce Web Sites by Considering Customer Reviews

Method:

Proposed an unsupervised framework using a Hidden Conditional Random Field (HCRF) to find popular product attributes from product description pages. • Used LDA (Latent Dirichlet Allocation) on customer reviews to discover popular product features and hidden topics. • Used

two hidden variable chains to identify attribute categories and mark the text containing product attributes.

Dataset:

Data collected from more than 20 e-commerce websites covering 13 product domains. • Used 661 product description pages in total. • Examples include baby car seats, digital cameras, GPS devices, LCD TVs, and netbooks. • Used more than 500 customer reviews for each product domain.

13.Enhancing the Efficiency of Sentiment Analysis for Vietnamese Service Reviews.

Method:

Proposed three methods for improving Vietnamese sentiment analysis. • Title + Content: Used both the review title and main content through separate deep learning branches and combined their features before classification. • VED_PhBERT: Created a Vietnamese Emoji Dictionary with 3,790 emoji entries and converted emojis into Vietnamese text before using PhoBERT. • E2V_PhBERT: Converted emojis into 300-dimensional Emoji2Vec vectors, changed them into 768-dimensional vectors, and combined them with PhoBERT embeddings.

Dataset:

Hotel Review Dataset: 8,404 Vietnamese hotel reviews from online platforms. Reviews were divided into Positive (4–5 stars), Neutral (3 stars), and Negative (1–2 stars). • Also tested the methods on three Vietnamese NLP datasets: UIT-ViHSD, UIT-VSFC, and AIVIVN 2019.

14.The Argument Reasoning Comprehension Task: Identification and Reconstruction of Implicit Warrants

Dataset:

Source: 5,000 comments from the New York Times Room for Debate. • Created 1,970 benchmark instances for testing implicit reasoning.

Method:

Created an 8-step crowdsourcing process to identify and validate implicit warrants. • Changed argument positions to create different reasoning cases. • Tested neural attention models and language models. • Compared AI performance with human performance.

15.The Argument Reasoning Comprehension Task: Identification and Reconstruction of Implicit Warrants.

Dataset:

Source: 5,000 comments from The New York Times: Room for Debate. • Created 1,970 high-quality benchmark examples for testing argument reasoning.

Method:

Created an 8-step crowdsourcing process to identify and check implicit warrants. • Changed argument positions to create alternative reasoning paths. • Tested neural attention models and language models. • Compared AI performance with human performance.

16.

Dataset:

• Source: Amazon.com digital camera reviews. • 214 reviews containing about 3,000 sentences. • Covered 6 camera models and 8 product features. • Used 368 user queries collected from 46 participants.

Method:

Used the Vector Space Model to identify relevant opinion sentences. • Proposed Collaborative-Filtering-based Query Expansion (CF-OSI). • Used Jaccard similarity to find similar previous user queries. • Expanded short queries using related user queries. • Used Cosine similarity to retrieve the most relevant opinion sentences.

17. Show Me the Money! Deriving the Pricing Power of Product Features by Mining Consumer Reviews.

Method:

Combined text mining / NLP with econometric analysis. • Extracted product features and opinion words from customer reviews. • Used a Tensor Product Space to connect product features with customer evaluations. • Used a low-dimensional Rank-1 approximation to reduce model complexity. • Studied how review information affects product demand and sales.

Knowledge Empowered Prominent Aspect Extraction from Product Reviews.

Dataset:

Used customer reviews from 6 product/service domains. • Hotel: 3.1 million reviews (20% sample). • Mobile Phone: 185.9K reviews. • MP3 Player: 31K reviews. • Laptop: 40.7K reviews. • Cameras: 471.1K reviews. • Restaurant: 269K reviews. • Tested with 5, 7, and 9 prominent aspects.

Method:

Proposed the ExtRA framework for extracting important product aspects. • Used lexico-syntactic rules to find aspect candidates. • Used WordNet and GloVe to group similar terms. • Used Probase to understand relationships between concepts. • Used K-means clustering with Silhouette score to create aspect groups. • Used Top-down, Bottom-up, and Miscellaneous strategies to select important, non-overlapping aspects.

